

Project Report

Date	28-06-2026
Project Name	Software Installation Request Automation
Team ID	
Maximum Marks	

Project Title: Software Installation Request Automation in ServiceNow

Project Overview

The Software Installation Request Automation project in ServiceNow is designed to streamline and automate the end-to-end lifecycle of software-related requests. Traditionally, organizations handle software requests through manual coordination and unstructured channels, which often results in severe delays, human errors, and limited visibility into the request status.

This project leverages the ServiceNow platform to resolve these inefficiencies by automating the complete request lifecycle. End-users can easily submit their requirements through a centralized, user-friendly Service Catalog rather than relying on manual emails. Once a request is submitted, the system's automated workflows take over to:

- Validate the request using dynamic Catalog Variables and UI Policies to ensure no incomplete data is submitted.
- Trigger dynamic approval processes, routing the request to Managers or Security Officers to enforce compliance with licensing and IT governance policies.
- Automatically create backend records across core tables (Request, Requested Item, and Catalog Task) and assign fulfillment tasks directly to the Software Support Team.
- Execute Business Rules to handle edge cases, such as automatically creating an incident if a requested software license is unavailable.
- Send real-time, automated email notifications to the requester at the submission, approval, and completion stages to eliminate the anxiety of an IT "black hole".

Ultimately, by integrating these ServiceNow capabilities (Service Catalog, UI Policies, Business Rules, and Workflow Editor), the solution provides a scalable and efficient architecture. It significantly reduces the administrative burden on IT staff, ensures better SLA compliance, and greatly enhances the end-user experience through faster, transparent service delivery.

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1. Introduction

Modern organizations depend heavily on equipping their employees with the right software tools for daily business operations. Employees regularly submit requests for various software applications, and managing these requests manually consumes significant administrative effort while increasing the likelihood of delays and errors. As organizations grow, manual request handling makes it difficult to monitor request progress, enforce consistent approval procedures, and ensure strict compliance with licensing and security policies.

The Software Installation Request Automation project in ServiceNow addresses these challenges by implementing a centralized, automated request management system. The solution enables end-users to submit their requirements through a user-friendly Service Catalog, moving away from unstructured manual coordination. Once submitted, the ServiceNow platform takes over to automate request validation, route approvals dynamically to Managers or Security Officers, and generate fulfillment tasks for the IT Support team.

The system provides:

- Faster software access and request processing.
- Standardized workflows and dynamic approval processes.
- Reduced manual effort and elimination of delays in request handling.
- Improved visibility and transparent tracking through a centralized platform.
- Better compliance with licensing, IT governance, and security policies.

Using ServiceNow's powerful workflow and automation capabilities, this project demonstrates how IT Service Management (ITSM) can automate repetitive operational tasks, effectively map business requirements to core database tables (REQ, RITM, SCTASK), and significantly improve the overall end-user experience.

2. Project Objectives

The primary objective of this project is to streamline and automate the end-to-end lifecycle of software installation requests within the ServiceNow platform, minimizing manual intervention and accelerating fulfillment times.

The project aims to provide a structured environment where users can easily submit requests through a Service Catalog, while the platform automatically handles approvals, notifications, record creation, and workflow execution.

Specific objectives include:

- Develop a centralized Service Catalog item for software requests to replace unstructured manual coordination.
- Reduce manual effort and eliminate delays in request handling.
- Accelerate approval and fulfillment times for software installations.
- Enforce standardized workflows and dynamic approval processes to ensure compliance with licensing, IT governance, and security policies.
- Improve visibility and tracking of requests through a centralized platform, providing automated notifications for submission, approval, and completion.
- Enhance the end-user experience by providing faster, transparent service delivery.
- Automatically generate and map backend records (Request, Requested Item, and Catalog Task) to assign fulfillment tasks directly to the IT support team.

3. Key Features

The proposed system includes the following major features:

- **User-Friendly Service Catalog:** A centralized, easy-to-use interface for employees to submit software installation requests, eliminating the need for unstructured emails.
- **Dynamic Forms & Catalog Variables:** Intelligent forms that capture specific request details using configured variables such as Software Name, Version Required, License Justification, and Urgency.
- **Data Integrity & Validation:** Implementation of Catalog UI Policies and Business Rules to enforce mandatory fields and prevent incomplete submissions.
- **Automated Approval Workflows:** Utilization of the Workflow Editor to dynamically route requests to Managers or Security Officers for licensing and compliance validation.
- **Automatic Record Creation:** Seamless mapping and automatic generation of backend records across core ServiceNow tables, including Requests (sc_request), Requested Items (sc_req_item), and Catalog Tasks (sc_task).
- **Automated Task Assignment:** The system automatically generates and assigns fulfillment tasks directly to the IT support/Software team upon manager approval.
- **Custom Incident Logic:** A specialized Business Rule that automatically creates an incident ticket if a requested software license is unavailable.
- **Real-Time Email Notifications:** Automated email triggers sent to users during critical stages (submission, approval, and completion) to eliminate status-chasing.
- **Transparent Request Tracking:** Real-time visibility into the order status and estimated delivery dates directly through the Service Portal.
- **Reduced Manual Effort:** Minimizes administrative overhead and manual coordination, leading to accelerated fulfillment times and better SLA compliance.

4. Project Milestones

The project implementation follows six major milestones.

Milestone 1: Update Sets & Project Setup

- **Objective:** Establish a controlled environment to capture, package, and migrate all customizations.
- **Activities:**
 - Identify the scope of changes (Business Rules, UI Policies, Workflows, etc.).
 - Define naming conventions and documentation standards for the update set.
 - Create a new Update Set in the development environment before beginning configuration.

Milestone 2: Service Catalog & Form Creation

- **Objective:** Build a structured, user-friendly entry point for users to request software installations.
- **Activities:**
 - Create the "Software Installation Request" catalog item under the appropriate category.
 - Configure dynamic Catalog Variables including Software Name, Version Required, License Justification, and Urgency.
 - Implement UI Policies and validations to ensure forms are user-friendly and prevent incomplete submissions.

Milestone 3: Backend Configuration & Data Architecture

- **Objective:** Define how request data will be captured, mapped, and managed in the ServiceNow database.
- **Activities:**
 - Identify and configure the Request-related tables: Request (sc_request), Requested Item (sc_req_item), and Catalog Task (sc_task).
 - Determine the relationships between the tables (REQ → RITM → SCTASK).
 - Develop Server-side Business Rules, such as automatically creating an Incident if a requested software license is unavailable.

Milestone 4: Workflow Design & Automation Integration

- **Objective:** Define and build a clear, automated structure for routing approvals and assigning tasks.
- **Activities:**

- Use the Workflow Editor to build the "Software Installation Request Workflow".
- Configure dynamic approval routing to Managers or Security Officers for compliance enforcement.
- Automate the generation of Catalog Tasks and assign them directly to the IT Support/Software Team.
- Attach the completed workflow to the Service Catalog item.

Milestone 5: Notifications & Real-Time Tracking

- **Objective:** Ensure complete transparency and eliminate status-chasing through automated communication.
- **Activities:**
 - Configure automated email notifications to trigger upon request submission.
 - Set up alerts for approval/rejection behaviors.
 - Configure a final completion notification once the IT fulfiller resolves the task.

Milestone 6: Testing, Validation, & Deployment

- **Objective:** Ensure the automated process works seamlessly from submission to fulfillment before migrating to production.
- **Activities:**
 - Submit test requests via the Service Portal to confirm UI policies and variable mappings.
 - Validate that approval behaviors and task auto-assignments function properly.
 - Change the Update Set state to "Complete" and migrate configurations from Development to Test/Production environments.

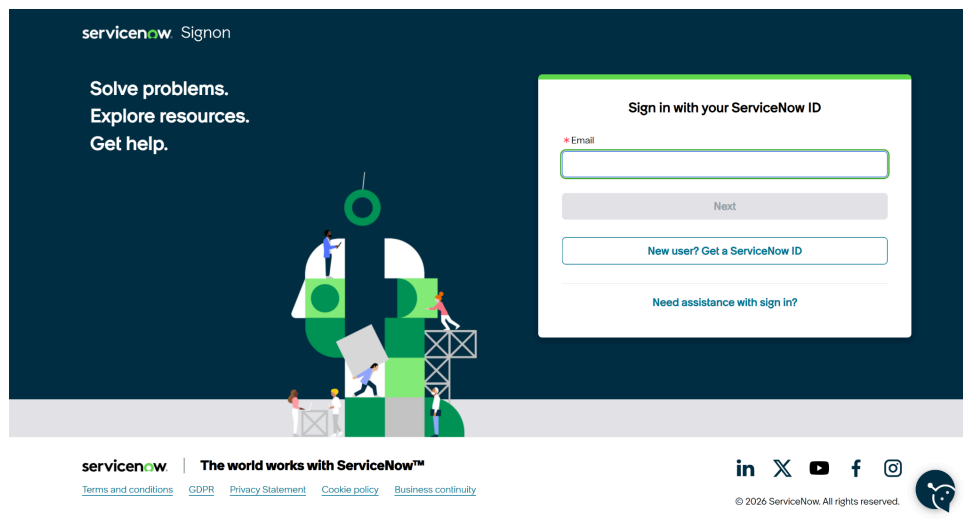
5. ServiceNow Developer Setup

Before developing the application, a ServiceNow Personal Developer Instance (PDI) must be created. The PDI provides a fully functional ServiceNow environment where the catalog items, update sets, and workflows can be developed, tested, and deployed safely without affecting production systems.

5.1 Creating a ServiceNow Developer Account

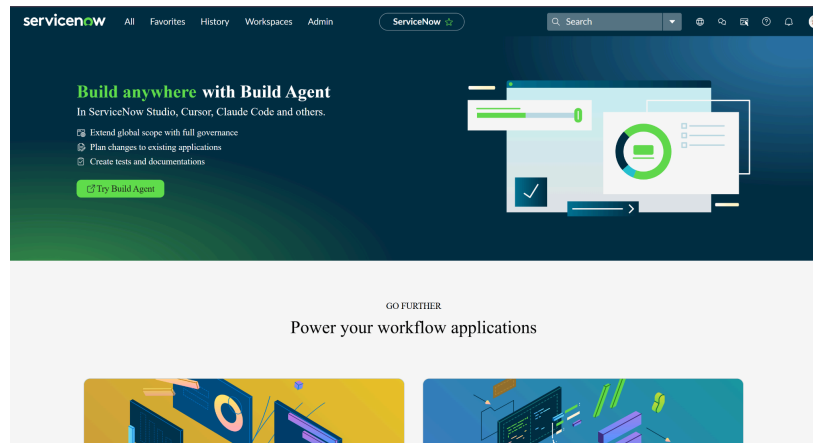
To begin development, perform the following steps:

- Visit the ServiceNow Developer Portal.
- Click Sign Up to create a free developer account.
- Enter the required information, including First Name, Last Name, Email Address, Country, and Password.
- Verify the email address using the activation link sent to your registered email.
- Log in to the ServiceNow Developer Portal.
- Click Start Building and request a Personal Developer Instance (PDI).
- Wait for the instance to be provisioned, then launch the instance and log in using the generated credentials.



5.2 Accessing the Development Environment Once the Personal Developer Instance is ready, the developer can access the core ServiceNow environments and tools required to build the automation application. By using the Application Navigator, the developer accesses the low-code and no-code tools utilized in this project, which include:

- **Maintain Items / Catalog Builder:** To create the Service Catalog forms, Catalog Variables, and UI Policies.
- **Workflow Editor:** To construct the visual flow logic for approvals, notifications, and task assignments.
- **Update Sets:** To capture all configuration changes for safe migration to production environments.
- **System Definition:** To define custom Business Rules and manage data architecture.



6. Project Implementation in ServiceNow

The implementation of the Software Installation Request Automation project consists of multiple interconnected modules within the ServiceNow platform. Each module contributes to automating the complete request lifecycle—from the moment the end-user accesses the portal to the final software installation by the IT support team.

The implementation was executed in a structured sequence: Update Set configuration, Service Catalog creation, Variable and UI Policy setup, Server-side Business Rules, Workflow automation, and final deployment.

6.1 Initial Setup: Update Sets Configuration

Before making any configuration changes, an Update Set was established to capture, package, and migrate all customizations from the development environment to the production environment.

Implementation Steps

- Navigated to **Local Update Sets** in the Application Navigator.
- Created a new Update Set named "**Software Installation Request Setup**".
- Set the state to "**In Progress**" and made it the current active update set to ensure all subsequent changes (Business Rules, UI Policies, Workflows, and Catalog Items) were accurately captured for deployment.

The screenshot shows the ServiceNow interface for configuring an Update Set. The breadcrumb trail is 'Update Set - Software Installation Request Automation'. The main form contains the following fields:

- Name:** Software Installation Request Automation
- State:** In progress (selected from a dropdown menu)
- Parent:** (empty field with a search icon)
- Release date:** (empty field with a calendar icon)
- Application:** Global
- Created:** 2026-06-28 01:35:08
- Created by:** admin
- Merged to:** (empty field)

Below these fields, there is a section for 'The source system for this update set' with an 'Installed from' field and a 'Description' field.

At the bottom left, there are 'Update' and 'Back Out' buttons, and a 'Related Links' section with links to 'Export to XML', 'Merge With Another Update Set', and 'Scan Update Set'.

6.2 Service Catalog Creation

The Service Catalog provides the self-service interface where employees can submit their software requirements. This replaces manual, unstructured email requests with a standardized form.

Implementation Steps

- Navigated to **Service Catalog** → **Catalog Definitions** → **Maintain Items**.
- Clicked **New** to create a new Catalog Item.
- Configured the following details:
 - **Name:** Software Installation Request
 - **Catalog:** Service Catalog
 - **Category:** Software
 - **Short Description:** Request to install company-approved software.
 - **Description:** Allows employees to request software installations. Fulfills via Software Support Team.
- Added a **Meta** tag (e.g., "New Software") to ensure the item is easily searchable by end-users in the Service Portal.

The screenshot shows the ServiceNow interface for creating a new Catalog Item. The breadcrumb trail is 'Service Catalog' → 'Catalog Definitions' → 'Maintain Items'. The page title is 'Catalog Item - Software Installation Request'. A blue banner at the top says 'Build and modify items faster with the improved Catalog Builder.' Below this, a light blue box contains instructions: 'Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies. • Enter a Name and Short description to display for the item. • Enter a Price, approvals, variables, and other information as needed.' The form fields are as follows: 'Name' is 'Software Installation Request'; 'Application' is 'Global'; 'Catalogs' is 'Service Catalog'; 'Category' is 'Software'; 'State' is 'None --'; 'Checked out' is 'None --'; 'Owner' is 'System Administrator'; 'Meta' is 'New Software'; 'Active' is checked; 'Fulfillment automation level' is 'Unspecified'. At the bottom, there are tabs for 'Item Details', 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings'. The 'Short description' field is 'Software Installation Request'.

6.3 Catalog Variables Configuration

To eliminate incomplete requests, Catalog Variables were created to collect specific, structured information from the requester.

Implementation Steps

- From the Catalog Item form, navigated to the **Variables** related list and clicked **New**.
- Created the following sequential variables:

1. Software Name

- **Type:** Single Line Text
- **Question:** What software do you need?

2. Version Required

- **Type:** Single Line Text
- **Question:** Specify version (if required).

3. License Justification

- **Type:** Multi Line Text
- **Question:** Why do you need this software?

4. Urgency

- **Type:** Choice
- **Question:** Select urgency level.
- **Choices Added:**
 - Normal
 - High
 - Critical

Variables (4)

Variable Sets

Catalog UI Policies

Catalog Client Scripts

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

≡

▼

Order

Search

⌕

—

Actions on selected rows...

New

Catalog Item = Software Installation Request

☐ 🔍 Type	Question	Order ▲
Multiple Choice	Select urgency level..	
Single Line Text	What Software do you need?	
Single Line Text	Specify version(if required)..	
Multi Line Text	Why do you need this software?	

◀◀

◀

1 to 4 of 4

▶

▶▶

Question Choices			
Order			
Search			
Question = Select urgency level..			
Actions on selected rows...			
New			
Text	Value	Order	Inactive
Normal	Normal	100	false
High	High	200	false
Critical	Critical	300	false
Insert a new row...			
1 to 3 of 3			

6.4 Data Validation and UI Configuration

To ensure data integrity, mandatory field enforcement was applied. Software-related details such as Software Name, Version, and Business Justification were made mandatory at the client side to prevent users from submitting blank or incomplete forms. Tooltips and Help Text icons were also added to assist users with specific fields, such as checking if they already hold a license.

The screenshot displays the 'Software Installation Request' form within the ServiceNow interface. The breadcrumb trail at the top reads: Home > Service Catalog > Software > Software Installation Request. A search bar labeled 'Search Catalog' is located in the top right corner. The main form area is titled 'Software Installation Request' and includes a subtitle 'Software Installation Request'. It features a 'Select urgency level..' section with three radio buttons: 'Normal' (selected), 'High', and 'Critical'. Below this is a text field for 'What Software do you need?'. Another text field is labeled 'Specify version(if required)..'. A third text field is labeled 'Why do you need this software?' with a red question mark icon. At the bottom, there is an 'Add attachments' section with a dashed border and a cloud icon, containing the text 'Choose a file or drag it here.' and 'Copy and paste clipboard files here.'. To the right of the main form is a sidebar with a 'Quantity:' dropdown set to '1', a 'Delivery Time: 0 Days' label, and three buttons: 'Add to Cart' (with a shopping cart icon), 'Save as Draft' (with a document icon), and 'Order Now' (a prominent blue button).

6.5 Backend Data Architecture & Table Handling

The project utilizes ServiceNow's core ITSM tables to capture and manage the catalog item requests. The data architecture was mapped to ensure seamless flow from the user to the IT team.

Table Relationships Established

- **Request (sc_request):** The parent record tracking the overall order.
- **Requested Item (sc_req_item):** The specific "Software Installation Request" item bound to the parent REQ, which houses the user's variables.
- **Catalog Task (sc_task):** The actual fulfillment task assigned to the network/software team to perform the installation.

Home

>

Request Summary - REQ0010016

Search Catalog

Q

Submitted :2026-06-28 08:24:44

Request Number : **REQ0010016**

Estimated Delivery : 2026-06-28

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Software Installation Request	2026-06-28		---	1	---
					Total: \$0.00

6.6 Server-Side Automation (Business Rules)

To handle complex backend logic and licensing compliance, a Business Rule was implemented. This script automatically triggers an Incident if the requested software cannot be installed due to a lack of available licenses.

Implementation Steps

- Navigated to **System Definition** → **Business Rules** and clicked **New**.
- **Name:** Create Incident on License Issue.
- **Table:** Requested Item (`sc_req_item`).
- **Advanced:** Checked the **Advanced** box.
- **When to Run:** Set to **After**, on **Update**.
- **Filter Conditions:** State changes to **"On Hold"** (or a custom field like Issue = True).
- **Script Configuration:** Wrote a GlideRecord script to automatically generate the incident.

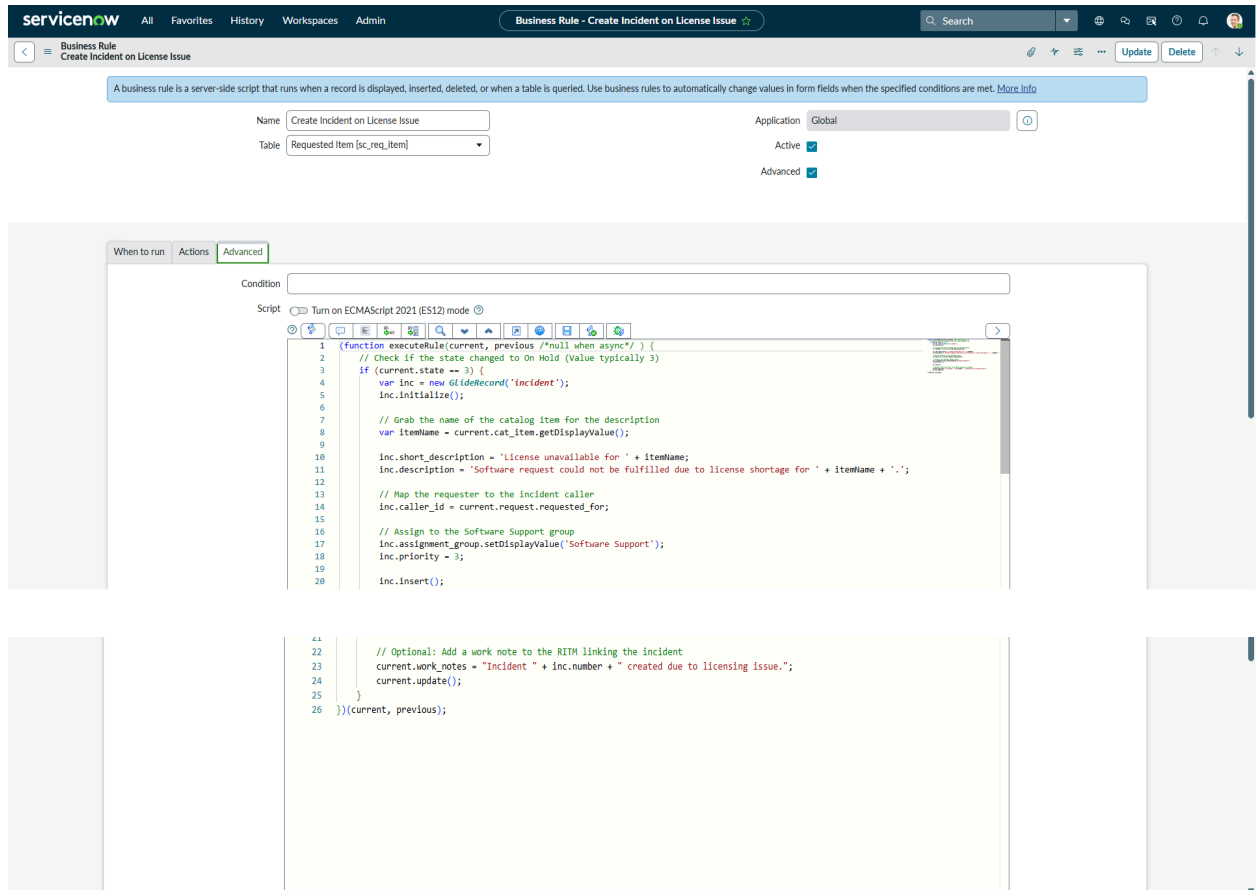
Business Rule Script

```
if (current.state == 3) { // On Hold
  var inc = new GlideRecord('incident');
  inc.initialize();
  inc.short_description = 'License unavailable for ' + current.request_item;
  inc.description = 'Software request could not be fulfilled due to license shortage.';
  inc.caller_id = current.request.requested_for;
  inc.assignment_group.setDisplayValue('Software Support');
```

```

inc.priority = 3;
inc.insert();
}

```



6.7 Workflow Editor Implementation

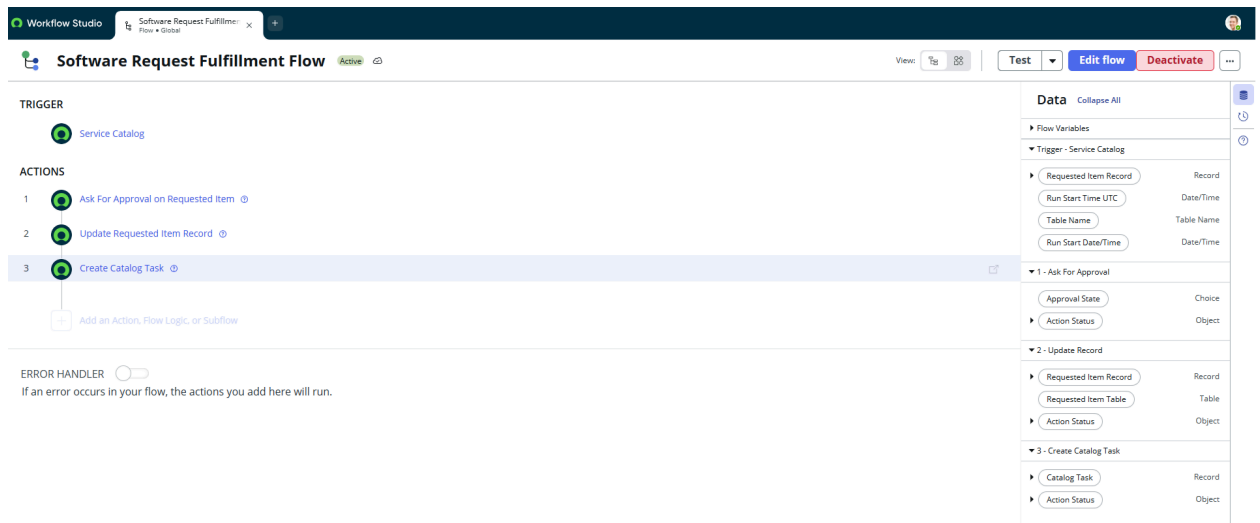
The Workflow Editor is the core automation engine of the project. It orchestrates dynamic approvals, automated task creation, and seamless fulfillment tracking for the software requests.

6.7.1 Workflow Setup

Implementation Steps

- Navigated to **Workflow** → **Workflow Editor** and clicked **New Workflow**.
- **Name:** Software Request Workflow.

- **Table:** Requested Item (sc_req_item).
- **Description:** Handles software request approvals and fulfillment tasks.



6.7.2 Approval Routing & Task Generation

- **Approval Step:** Dragged the **Approval - Group** (or Approval - User) core activity onto the canvas to route the request to Managers/Security Officers for compliance validation.
- **Task Step:** Dragged the **Catalog Task** activity to the canvas to automatically assign the fulfillment work to the **Software Support** team.
- **End Step:** Connected the final node to the **End** activity to mark workflow completion.
- **Publishing:** Connected all activities in logical sequence (**Begin** → **Approval** → **Task** → **End**) and clicked **Publish**.

6.7.3 Attaching Workflow to Catalog Item

- Returned to the **Software Installation Request** Catalog Item.
- Navigated to the **Process Engine** tab.
- In the **Workflow** field, selected the newly published **Software Request Workflow**.
- Saved the record to bind the automation logic to the user-facing form.

Build and modify items faster with the improved Catalog Builder.

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name: Software Installation Request

Application: Global

Catalogs: Service Catalog

Category: Software

State: -- None --

Checked out: -- None --

Owner: System Administrator

Meta: New Software

Active: ☒

Fulfillment automation level: Unspecified

Item Details | **Process Engine** | Picture | Pricing | Portal Settings

Select the appropriate process engine for the catalog item. Only one engine can be selected.

Flow: Software Request Fulfillment Flow

Workflow:

Execution Plan:

6.8 Service Portal Integration & User Experience

To validate the end-user experience, the system was tested in the Service Portal environment (/sp).

Implementation & Testing Steps

- Copied the instance domain and appended **/sp** (e.g., <https://devXXXXX.service-now.com/sp>).
- Searched for **"New Software"** utilizing the previously configured meta tags.
- Filled out the form variables and clicked **Order Now** to initiate the request.
- Verified the **Order Status** page, which provided the user with immediate visibility into the Request Number (REQ), Estimated Delivery Date, Stage, and Price.

6.9 Update Set Migration & Deployment

The final step of the implementation phase was deploying the completed automation into the production/test environment.

Implementation Steps

- Navigated back to **Local Update Sets** and opened **"Software Installation Request Setup"**.
- Changed the State from **"In Progress"** to **"Complete"** and saved.

- Clicked the **Export to XML** related link to download the configuration payload.
- Logged into the target ServiceNow instance and navigated to **Retrieved Update Sets**.
- Clicked **Import Update Set from XML**, uploaded the file, and clicked **Preview Update Set** to check for collisions.
- Once previewed successfully, clicked **Commit Update Set** to transfer the entire data model, workflow, catalog item, and scripts to the new environment.
- Validated the **Customer Updates** related list to ensure all variables, workflows, and policies were successfully inserted or updated.

The screenshot shows the ServiceNow interface for configuring an Update Set. The main form includes the following fields:

- Name:** Software Installation Request Automation
- State:** Complete
- Parent:** (empty)
- Release date:** (empty)
- Install date:** (empty)
- Installed from:** (empty)
- Description:** (empty text area)

Metadata fields on the right include:

- Application:** Global
- Created:** 2026-06-28 01:35:08
- Created by:** admin
- Merged to:** (empty)

At the bottom left, there are buttons for **Update** and **Back Out**. Below these are the **Related Links**:

- [Export to XML](#)
- [Merge With Another Update Set](#)
- [Scan Update Set](#)

7. Conclusion

The Software Installation Request Automation project successfully demonstrates how the ServiceNow platform can be leveraged to streamline and automate IT service delivery. By moving away from manual, unstructured email requests, this project established a structured, efficient process for managing software requests from initial submission to final completion.

Through the integration of the Service Catalog, dynamic Catalog Variables, and Client-side UI Policies, the system ensures that end-users experience a clean, intuitive interface that enforces data integrity by preventing incomplete submissions. The implementation of the Workflow Editor orchestrates dynamic approval routing, ensuring that licensing and security policies are strictly enforced before automatically generating fulfillment tasks for the IT Support team. Furthermore, custom Business Rules handle complex edge cases, such as the automatic creation of incidents during license shortages.

Overall, this automation significantly reduces the manual administrative burden on IT staff, minimizes processing delays, and improves SLA compliance. By providing automated email notifications and real-time tracking via the Service Portal, the solution eliminates the frustration of the IT "black hole," delivering a highly scalable, user-centric system that vastly improves organizational efficiency and user satisfaction.

